

BAND-LIMITEDNESS OF THE HARDY Z -FUNCTION IN THE PHASE VARIABLE AND ITS HIGHER CUMULANT SPECTRA

STEPHEN CROWLEY

ABSTRACT. Let $\vartheta(t) = \operatorname{Im} \log \Gamma\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{t}{2} \log \pi$ be the Riemann–Siegel theta function and $Z(t) = e^{i\vartheta(t)} \zeta\left(\frac{1}{2} + it\right)$ the Hardy Z -function. Fix any constant $C > c_*$, where

$$c_* := \frac{1}{2}(\gamma + 3 \log 2 + \frac{\pi}{2} + \log \pi) \approx 2.6860917,$$

so that $\Theta(t) := \vartheta(t) + Ct$ is a C^∞ diffeomorphism of $[0, \infty)$, and let $\tilde{Z} := Z \circ \Theta^{-1}$. We prove that for every $\xi \in \mathbb{R}$ with $|\xi| > 1$,

$$\frac{1}{\Theta(T)} \int_0^{\Theta(T)} \tilde{Z}(u) e^{-i\xi u} du \longrightarrow 0 \quad \text{as } T \rightarrow \infty,$$

so the Cesàro spectral support of $\tilde{Z}$ is contained in $[-1, 1]$. The proof rests on three ingredients: (i) the Riemann–Siegel expansion with remainder $O(t^{-1/4})$; (ii) a single integration by parts whose tail is eliminated by an exact antiderivative; and (iii) the elementary bound $\omega_n(u) \in [0, 1)$, where u_n is the exact saddle of ω_n . The cutoff $|\xi| = 1$ is sharp. We then extend the argument to higher-order cumulant spectra: for every $s \geq 2$ the s -point polyspectrum Φ_s of $\tilde{Z}$ is supported in the box $[-1, 1]^{s-1}$.

CONTENTS

1. Introduction	1
2. Setup and Notation	2
3. Riemann–Siegel Expansion	3
4. Phase-Derivative Identity	3
5. The Uniform IBP Bound	4
6. Band-Limitedness	5
7. Sharpness	6
8. Higher Cumulant Spectra	7
8.1. Cumulant densities and polyspectra	8
8.2. Multi-mode IBP	8
8.3. Band-limitedness of Φ_s	9
Acknowledgements	10
References	10

Date: April 29, 2026.

2020 Mathematics Subject Classification. 11M06, 11M26, 42A38, 60G10, 62M15.

Key words and phrases. Hardy Z -function, Riemann–Siegel formula, band-limited function, phase variable, Cesàro–Fourier transform, polyspectrum, cumulant.

1. INTRODUCTION

The Riemann–Siegel theta function $\vartheta(t)$ satisfies $\vartheta'(t) \sim \frac{1}{2} \log(t/2\pi) \rightarrow +\infty$, so the Hardy Z -function oscillates at a rate that grows without bound. A natural strategy is to absorb this growth into a change of variables: replace the time parameter t by the *accumulated phase* $u = \Theta(t) := \vartheta(t) + Ct$, where $C > c_*$ is chosen so that Θ is a globally increasing diffeomorphism. The reparametrized function $\tilde{Z}(u) := Z(\Theta^{-1}(u))$ then admits a well-defined spectral theory in the Cesàro sense.

The main result, Theorem 6.1, shows that the Cesàro–Fourier transform of $\tilde{Z}$ is supported in $[-1, 1]$. The argument proceeds in three steps that mirror the Riemann–Siegel decomposition (1). The cosine sum contributes pairs of mode integrals $I_n(\xi, T)$ and $J_n(\xi, T)$, each bounded via a single integration by parts (Lemmas 5.1–5.2); the error term is absorbed by the classical $O(t^{-1/4})$ bound on the Riemann–Siegel remainder.

A distinguishing feature of this argument is that the tail integral in the integration by parts telescopes *exactly*, yielding the sharp bound $|I_n(\xi, T)| \leq 2/(\xi - 1)$ with no asymptotic error. The cutoff $|\xi| = 1$ is confirmed to be optimal in Corollary 7.1: at every interior frequency $\xi \in (-1, 1)$ the phase has a non-degenerate stationary point, and after a mode-scale separation argument the Cesàro average remains bounded below.

Higher cumulant spectra. Section 8 extends these results to the s -point *polyspectrum* Φ_s of $\tilde{Z}$, i.e., the Fourier transform of the s -point Cesàro cumulant density κ_s . We show that Φ_s is supported in $[-1, 1]^{s-1}$ (Theorem 8.5), generalizing the band-limitedness of the two-point case. The proof uses a multi-mode IBP in which each frequency variable ξ_j is handled independently via Lemma 5.1.

Corrections to a prior draft. This revision fixes three issues in an earlier version of the manuscript [9]. (i) The identity $\vartheta'(2\pi n^2) = \log n$ is only asymptotic: $\vartheta'(2\pi n^2) = \log n - 1/(16\pi^2 n^4) + O(n^{-8})$. We therefore redefine u_n as $\Theta(t_n^\dagger)$, where $t_n^\dagger$ is the *exact* solution of $\vartheta'(t) = \log n$, so that $\omega_n(u_n) = 0$ is an identity by construction. (ii) The identity $2 \cos \Phi_n = e^{i\Phi_n} + e^{-i\Phi_n}$ produces two distinct mode integrals I_n and J_n ; the claim “ $2 \operatorname{Re} I_n$ ” in the earlier draft conflates them. We bound both by separate (identical) IBPs. (iii) The mode-separation step in the sharpness proof is made explicit using the asymptotic scales $t_n^* \sim 2\pi n^{2/(1-\xi)} e^{2\xi C/(1-\xi)}$.

Notation. γ denotes the Euler–Mascheroni constant, $\psi = \Gamma'/\Gamma$ the digamma function, $f \ll g$ means $|f| \leq Cg$ for an absolute constant $C > 0$, and $\lfloor \cdot \rfloor$ is the floor function.

2. SETUP AND NOTATION

Definition 2.1 (Hardy Z -function). For $t \geq 0$ define

$$\vartheta(t) := \operatorname{Im} \log \Gamma\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{t}{2} \log \pi, \quad Z(t) := e^{i\vartheta(t)} \zeta\left(\frac{1}{2} + it\right).$$

The function Z is real-valued with zeros exactly the zeros of ζ on $\operatorname{Re} s = \frac{1}{2}$.

Definition 2.2 (Shifted phase map). Set

$$c_* := \frac{1}{2} \left(\gamma + 3 \log 2 + \frac{\pi}{2} + \log \pi \right),$$

fix any $C > c_*$, and define

$$\Theta(t) := \vartheta(t) + Ct, \quad g := \Theta^{-1}, \quad \tilde{Z}(u) := Z(g(u)).$$

Proposition 2.3 (Diffeomorphism property). *For any $C > c_*$, the map $\Theta : [0, \infty) \rightarrow [\Theta(0), \infty)$ is a C^∞ diffeomorphism with*

$$\Theta'(t) = \vartheta'(t) + C > 0 \quad \forall t \geq 0, \quad \Theta'(t) \sim \frac{1}{2} \log\left(\frac{t}{2\pi}\right) \quad (t \rightarrow \infty).$$

Moreover, $\Theta(T) \sim \frac{T}{2} \log(T/2\pi)$ as $T \rightarrow \infty$.

Proof. The identity $\vartheta'(t) = \frac{1}{2} \operatorname{Re} \psi\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{1}{2} \log \pi$ and the digamma special value $\psi\left(\frac{1}{4}\right) = -\gamma - \frac{\pi}{2} - 3 \log 2$ give

$$\vartheta'(0) = \frac{1}{2} \psi\left(\frac{1}{4}\right) - \frac{1}{2} \log \pi = -\frac{1}{2} \left(\gamma + \frac{\pi}{2} + 3 \log 2 + \log \pi \right) = -c_*.$$

Since ϑ' is strictly increasing with $\vartheta'(t) = \frac{1}{2} \log(t/2\pi) + O(t^{-2})$ [1], its global infimum is $-c_*$, so $\Theta'(t) = \vartheta'(t) + C > 0$ for all $t \geq 0$ whenever $C > c_*$. Surjectivity onto $[\Theta(0), \infty)$ is immediate from $\Theta(t) \rightarrow \infty$. The asymptotic for $\Theta(T)$ follows by integration. $\square$

3. RIEMANN–SIEGEL EXPANSION

Set $N(t) := \lfloor \sqrt{t/(2\pi)} \rfloor$. The Riemann–Siegel formula [2, 1] asserts

$$(1) \quad Z(t) = 2 \sum_{n=1}^{N(t)} n^{-1/2} \cos(\vartheta(t) - t \log n) + R(t), \quad |R(t)| \ll t^{-1/4}, \quad t \geq 1.$$

Lemma 3.1 (Partial-sum bound). *For all $T \geq 1$,*

$$\sum_{n=1}^{N(T)} n^{-1/2} \leq 2 \left(\frac{T}{2\pi} \right)^{1/4} + 2.$$

Proof. The standard estimate $\sum_{n=1}^M n^{-1/2} \leq 2\sqrt{M} + 1$, applied with $M = N(T) \leq (T/2\pi)^{1/2}$, gives the result. $\square$

4. PHASE-DERIVATIVE IDENTITY

Definition 4.1 (Exact saddle points). For each $n \geq 1$ let $t_n^\dagger$ denote the unique solution in $(0, \infty)$ of

$$\vartheta'(t_n^\dagger) = \log n,$$

and set $u_n := \Theta(t_n^\dagger)$. By the asymptotic $\vartheta'(t) = \frac{1}{2} \log(t/2\pi) + O(t^{-2})$,

$$t_n^\dagger = 2\pi n^2 + O(n^{-2}), \quad u_n = \Theta(2\pi n^2) + O\left(\frac{\log n}{n^2}\right).$$

Remark 4.2. An earlier draft [9] used $u_n = \Theta(2\pi n^2)$ directly, relying on the *asymptotic* identity $\vartheta'(2\pi n^2) = \log n$. The correction is of order n^{-4} : $\vartheta'(2\pi n^2) = \log n - \frac{1}{16\pi^2 n^4} + O(n^{-8})$. Using $t_n^\dagger$ in place of $2\pi n^2$ makes the saddle condition $\omega_n(u_n) = 0$ an exact identity, not an asymptotic one, at the negligible cost of an $O(n^{-4})$ perturbation of the lower limit of the mode integral.

For $n \geq 1$ and $u \geq u_n$, define the *mode phase* and *frequency ratio*

$$(2) \quad \Phi_n(u) := \vartheta(g(u)) - g(u) \log n, \quad \omega_n(u) := \frac{\vartheta'(g(u)) - \log n}{\Theta'(g(u))}.$$

Lemma 4.3 (Phase-derivative identity). *For all integers $n \geq 1$, all $u \geq u_n$, and all $\xi \in \mathbb{R}$:*

- (a) $\frac{d}{du}(\Phi_n(u) - \xi u) = \omega_n(u) - \xi$.
- (b) $\omega_n(u_n) = 0$ exactly, by Definition 4.1.
- (c) $0 \leq \omega_n(u) < 1$ for all $u \geq u_n$.
- (d) $\omega'_n(u) = \frac{(C + \log n) \vartheta''(g(u))}{\Theta'(g(u))^3} > 0$; hence ω_n is strictly increasing on $[u_n, \infty)$.
- (e) For every constant $\xi \notin \omega_n([u_n, \infty))$,

$$\frac{d}{du} \left[\frac{1}{\xi - \omega_n(u)} \right] = \frac{\omega'_n(u)}{(\xi - \omega_n(u))^2}.$$

Proof. (a). Using $g'(u) = 1/\Theta'(g(u))$:

$$\frac{d\Phi_n}{du} = (\vartheta'(g(u)) - \log n) \cdot g'(u) = \frac{\vartheta'(g(u)) - \log n}{\Theta'(g(u))} = \omega_n(u).$$

Subtracting ξ gives part (a).

(b). By Definition 4.1, $g(u_n) = t_n^\dagger$ and $\vartheta'(t_n^\dagger) = \log n$, so $\omega_n(u_n) = 0/\Theta'(t_n^\dagger) = 0$.

(c). Non-negativity: $\omega_n(u_n) = 0$ and ω_n is increasing by (d), so $\omega_n(u) \geq 0$. The upper bound follows from the algebraic identity

$$\omega_n(u) = 1 - \frac{\log n + C}{\vartheta'(g(u)) + C} < 1,$$

since $\log n + C > 0$ for all $n \geq 1$ and $C > 0$.

(d). Write $\omega_n = (f \circ g)/(h \circ g)$ with $f(t) := \vartheta'(t) - \log n$ and $h(t) := \Theta'(t) = \vartheta'(t) + C$. Then $f' = h' = \vartheta''$, and the quotient rule with $g' = 1/(h \circ g)$ gives

$$\begin{aligned} \omega'_n(u) &= \frac{f'(g) h(g) - f(g) h'(g)}{h(g)^2} \cdot g'(u) \\ &= \frac{\vartheta''(g) [h(g) - f(g)]}{h(g)^3} = \frac{(C + \log n) \vartheta''(g(u))}{\Theta'(g(u))^3}. \end{aligned}$$

Positivity $\vartheta''(t) > 0$ for all $t \geq 0$ is classical [3, 5.15.3].

(e). Direct computation: $\frac{d}{du}(\xi - \omega_n)^{-1} = \omega'_n(\xi - \omega_n)^{-2}$. □

5. THE UNIFORM IBP BOUND

For $\xi > 1$, $n \geq 1$, and T with $\Theta(T) > u_n$, define the *mode integrals*

$$(3) \quad I_n(\xi, T) := \int_{u_n}^{\Theta(T)} e^{i(\Phi_n(u) - \xi u)} du, \quad J_n(\xi, T) := \int_{u_n}^{\Theta(T)} e^{-i(\Phi_n(u) + \xi u)} du.$$

Lemma 5.1 (Uniform IBP bound for I_n). *For every $\xi > 1$, $n \geq 1$, and T with $\Theta(T) > u_n$,*

$$|I_n(\xi, T)| \leq \frac{2}{\xi - 1}.$$

Proof. Set $A := \xi - \omega_n(\Theta(T))$ and $B := \xi - \omega_n(u_n) = \xi$. By Lemma 4.3(c), $A \geq \xi - 1 > 0$ and $B = \xi > 1$.

Integration by parts. Write the integrand as $e^{i(\Phi_n - \xi u)} = (i(\omega_n - \xi))^{-1} \frac{d}{du} e^{i(\Phi_n - \xi u)}$ and integrate by parts once:

$$(4) \quad I_n(\xi, T) = \left[\frac{e^{i(\Phi_n - \xi u)}}{i(\omega_n - \xi)} \right]_{u_n}^{\Theta(T)} + \int_{u_n}^{\Theta(T)} \frac{e^{i(\Phi_n - \xi u)} \omega'_n(u)}{i(\xi - \omega_n(u))^2} du.$$

Boundary terms. The upper boundary contributes at most $1/A$; the lower contributes $1/B$.

Tail integral. Since $\omega'_n > 0$ by Lemma 4.3(d), the modulus of the tail integrand is non-negative, and Lemma 4.3(e) supplies the exact antiderivative:

$$\int_{u_n}^{\Theta(T)} \frac{\omega'_n(u)}{(\xi - \omega_n(u))^2} du = \left[\frac{1}{\xi - \omega_n(u)} \right]_{u_n}^{\Theta(T)} = \frac{1}{A} - \frac{1}{B}.$$

This is an identity, not an asymptotic.

Combining.

$$|I_n(\xi, T)| \leq \frac{1}{A} + \frac{1}{B} + \left(\frac{1}{A} - \frac{1}{B} \right) = \frac{2}{A} \leq \frac{2}{\xi - 1}. \quad \square$$

Lemma 5.2 (Uniform IBP bound for J_n). *For every $\xi > 1$, $n \geq 1$, and T with $\Theta(T) > u_n$,*

$$|J_n(\xi, T)| \leq \frac{2}{\xi} \leq \frac{2}{\xi - 1}.$$

Proof. The phase of the J_n integrand is $-(\Phi_n(u) + \xi u)$, with derivative $-(\omega_n(u) + \xi)$. By Lemma 4.3(c), $\omega_n(u) + \xi \geq \xi > 1$. Apply the same IBP as in Lemma 5.1, with $\xi - \omega_n$ replaced by $\xi + \omega_n$. Set $A' := \xi + \omega_n(\Theta(T))$ and $B' := \xi + \omega_n(u_n) = \xi$. Both $A' \geq \xi$ and $B' = \xi$. The boundary contributes $1/A' + 1/B' \leq 2/\xi$, and the tail antiderivative $-[(\xi + \omega_n)^{-1}]_{u_n}^{\Theta(T)} = 1/B' - 1/A'$ contributes at most $1/\xi - 1/A'$. Combining:

$$|J_n(\xi, T)| \leq \frac{1}{A'} + \frac{1}{B'} + \left(\frac{1}{B'} - \frac{1}{A'} \right) = \frac{2}{B'} = \frac{2}{\xi}. \quad \square$$

Remark 5.3. In both lemmas the $1/B$ (resp. $1/A'$) contributions cancel exactly between a boundary term and the tail antiderivative. No oscillatory phase cancellation is invoked.

6. BAND-LIMITEDNESS

Theorem 6.1 (Band-limitedness of $\tilde{Z}$). *Let $C > c_*$, $g = \Theta^{-1}$, and $\tilde{Z} = Z \circ g$. For every $\xi \in \mathbb{R}$ with $|\xi| > 1$,*

$$\frac{1}{\Theta(T)} \int_0^{\Theta(T)} \tilde{Z}(u) e^{-i\xi u} du \longrightarrow 0 \quad \text{as } T \rightarrow \infty.$$

Proof. Since Z is real-valued, the $\xi < -1$ case follows from $\xi > 1$ by complex conjugation. Fix $\xi > 1$ and set $U := \Theta(T)$.

Step 1 (Change of variables). Substituting $u = \Theta(t)$, $du = \Theta'(t) dt$, $g(u) = t$:

$$\int_0^U \tilde{Z}(u) e^{-i\xi u} du = \int_0^T Z(t) e^{-i\xi \Theta(t)} \Theta'(t) dt.$$

Insert the Riemann–Siegel expansion (1) and use $2 \cos \alpha = e^{i\alpha} + e^{-i\alpha}$:

$$Z(t) e^{-i\xi \Theta(t)} \Theta'(t) = \sum_{n=1}^{N(t)} n^{-1/2} [e^{i(\vartheta(t)-t \log n - \xi \Theta(t))} + e^{-i(\vartheta(t)-t \log n + \xi \Theta(t))}] \Theta'(t) + R(t) e^{-i\xi \Theta(t)} \Theta'(t).$$

Substituting $u = \Theta(t)$ converts the first bracketed term to $e^{i(\Phi_n(u) - \xi u)} du$ and the second to $e^{-i(\Phi_n(u) + \xi u)} du$, giving I_n and J_n respectively (the lower limit of integration shifts from 0 to u_n at the cost of an $O(n^{-1/2})$ boundary contribution absorbed into the $O(1)$ constants). Fubini's theorem applies since for each fixed T ,

$$\sum_{n \leq N(T)} n^{-1/2} (|I_n| + |J_n|) \ll \frac{T^{1/4}}{\xi - 1} < \infty$$

by Lemmas 5.1–5.2. Therefore

$$(5) \quad \int_0^U \tilde{Z}(u) e^{-i\xi u} du = \sum_{n=1}^{N(T)} n^{-1/2} (I_n(\xi, T) + J_n(\xi, T)) + S_R(\xi, T),$$

where $S_R(\xi, T) := \int_0^T R(t) e^{-i\xi \Theta(t)} \Theta'(t) dt$.

Step 2 (Cosine sum). By Lemmas 5.1, 5.2, and 3.1,

$$\left| \sum_{n=1}^{N(T)} n^{-1/2} (I_n(\xi, T) + J_n(\xi, T)) \right| \leq \frac{4}{\xi - 1} \left(2 \left(\frac{T}{2\pi} \right)^{1/4} + 2 \right) = O\left(\frac{T^{1/4}}{\xi - 1} \right).$$

Dividing by $\Theta(T) \sim \frac{T}{2} \log(T/2\pi)$ gives $O(T^{-3/4}/\log T) \rightarrow 0$.

Step 3 (Remainder). Using $|R(t)| \ll t^{-1/4}$ and $\Theta'(t) \ll \log t$,

$$|S_R(\xi, T)| \ll \int_1^T t^{-1/4} \log t dt \ll T^{3/4} \log T.$$

Dividing by $\Theta(T) \sim \frac{T}{2} \log T$ gives $O(T^{-1/4}) \rightarrow 0$.

Combining Steps 2 and 3 completes the proof. $\square$

7. SHARPNESS

Corollary 7.1 (Spectral support is exactly $[-1, 1]$). *For every $\xi \in (-1, 1) \setminus \{0\}$,*

$$\limsup_{T \rightarrow \infty} \frac{1}{\Theta(T)} \left| \int_0^{\Theta(T)} \tilde{Z}(u) e^{-i\xi u} du \right| > 0.$$

Consequently the Cesàro spectral support of $\tilde{Z}$ is exactly $[-1, 1]$.

Proof. Fix $\xi \in (0, 1)$; the case $\xi \in (-1, 0)$ follows by conjugation. For each $n \geq 1$, the equation $\omega_n(u) = \xi$ has the unique solution $u_n^* = \Theta(t_n^*)$, where t_n^* solves

$$\vartheta'(t_n^*) = \frac{\log n + \xi C}{1 - \xi}.$$

Inverting $\vartheta'(t) = \frac{1}{2} \log(t/2\pi) + O(t^{-2})$:

$$t_n^* \sim 2\pi n^{2/(1-\xi)} e^{2\xi C/(1-\xi)}.$$

Mode-scale separation. Consecutive scales satisfy $t_{n+1}^*/t_n^* = (1 + 1/n)^{2/(1-\xi)} \rightarrow 1^+$ with $t_{n+1}^* - t_n^* \sim (2/(1-\xi)) t_n^*/n \rightarrow \infty$, so the saddles are well-spaced on a logarithmic scale but not in absolute distance. Choose a subsequence T_n with $\Theta(T_n) \gg u_n^* + \delta$ for fixed $\delta > 0$.

Stationary-phase contribution at the n -th saddle. On a window $[u_n^* - \delta, u_n^* + \delta]$ the phase $\Phi_n(u) - \xi u$ has a non-degenerate critical point:

$$(\Phi_n - \xi u)''(u_n^*) = \omega'_n(u_n^*) > 0.$$

Truncating smoothly with a bump χ supported in that window, classical stationary phase [4] gives

$$\int \chi(u) e^{i(\Phi_n(u) - \xi u)} du = \sqrt{\frac{2\pi}{\omega'_n(u_n^*)}} e^{i(\Phi_n(u_n^*) - \xi u_n^*) + i\pi/4} + O((\omega'_n(u_n^*))^{-3/2}).$$

Since $\omega'_n(u_n^*) = (C + \log n) \vartheta''(t_n^*) / \Theta'(t_n^*)^3 \asymp 1/(t_n^* (\log t_n^*)^2)$, the leading amplitude scales as $\sqrt{t_n^*} \log t_n^*$.

Non-cancellation in the Cesàro sum. The contribution of the n -th mode to the Cesàro average at $T = T_n$ is

$$\frac{1}{\Theta(T_n)} \cdot n^{-1/2} \sqrt{\frac{2\pi}{\omega'_n(u_n^*)}} \asymp \frac{n^{-1/2} \sqrt{t_n^*} \log t_n^*}{t_n^* \log t_n^*} = \frac{n^{-1/2}}{\sqrt{t_n^*}} \asymp n^{-1/2-1/(1-\xi)}.$$

Because the saddles t_n^* are separated by gaps of order $t_n^*/n \rightarrow \infty$, contributions from distinct modes do not coherently cancel to leading order. Choosing $T = T_1$ and noting that the $n = 1$ contribution is $\asymp e^{-\xi C/(1-\xi)} > 0$ (and contributions from $n \geq 2$ are of

smaller amplitude at the T_1 scale), we obtain a strictly positive lower bound, giving $\limsup > 0$. $\square$

Remark 7.2. The spectral support is independent of the choice of $C > c_*$: replacing C by $C' > c_*$ rescales the t_n^* but preserves the non-vanishing of the Cesàro average. The leading amplitude for ξ near 0 scales as $e^{-\xi C/(1-\xi)}$, so the spectral density decays exponentially as $\xi \rightarrow \pm 1$ but is everywhere strictly positive on $(-1, 1) \setminus \{0\}$.

8. HIGHER CUMULANT SPECTRA

The band-limitedness result extends from the first-order spectral density to every higher cumulant. We introduce the s -point Cesàro cumulant density κ_s and its Fourier transform Φ_s (the s -point polyspectrum), and prove that Φ_s is supported in $[-1, 1]^{s-1}$.

8.1. Cumulant densities and polyspectra.

Definition 8.1 (s -point Cesàro moment and cumulant). For $s \geq 1$ and $\mathbf{v} = (v_1, \dots, v_{s-1}) \in \mathbb{R}^{s-1}$, define the s -point Cesàro moment density as the distributional limit

$$M_s(\mathbf{v}) := \lim_{T \rightarrow \infty} \frac{1}{\Theta(T)} \int_0^{\Theta(T)} \tilde{Z}(u) \prod_{j=1}^{s-1} \tilde{Z}(u + v_j) du,$$

assuming the limit exists in $\mathcal{S}'(\mathbb{R}^{s-1})$. The s -point cumulant density $\kappa_s(\mathbf{v})$ is obtained from $\{M_j\}_{j \leq s}$ by the Leonov–Shiryaev formula [5, 6]:

$$\kappa_s(\mathbf{v}) = \sum_{\pi \in \mathcal{P}(s)} (-1)^{|\pi|-1} (|\pi| - 1)! \prod_{B \in \pi} M_{|B|}(\mathbf{v}_B),$$

where $\mathcal{P}(s)$ is the lattice of set partitions of $\{0, 1, \dots, s-1\}$ (with $v_0 := 0$) and $\mathbf{v}_B = (v_j)_{j \in B}$.

Definition 8.2 (s -point polyspectrum). The s -point polyspectrum is the tempered distribution

$$\Phi_s(\boldsymbol{\xi}) := \int_{\mathbb{R}^{s-1}} \kappa_s(\mathbf{v}) e^{-i\boldsymbol{\xi} \cdot \mathbf{v}} d\mathbf{v}, \quad \boldsymbol{\xi} = (\xi_1, \dots, \xi_{s-1}) \in \mathbb{R}^{s-1}.$$

For $s = 2$ this is the ordinary spectral density of $\tilde{Z}$ (the squared modulus of its Cesàro–Fourier transform, up to normalization); for $s = 3$ this is the bispectrum; for $s = 4$ the trispectrum.

8.2. Multi-mode IBP. For $\mathbf{n} = (n_0, n_1, \dots, n_{s-1}) \in \mathbb{Z}_{\geq 1}^s$, $\boldsymbol{\epsilon} = (\epsilon_0, \dots, \epsilon_{s-1}) \in \{\pm 1\}^s$, and $\boldsymbol{\xi} \in \mathbb{R}^{s-1}$, define the *multi-mode integral*

$$(6) \quad I_{\mathbf{n}, \boldsymbol{\epsilon}}(\boldsymbol{\xi}, T) := \int_{[0, \Theta(T)] \times \mathbb{R}^{s-1}} \exp \left[i \sum_{j=0}^{s-1} \epsilon_j \Phi_{n_j}(u + v_j) - i \sum_{j=1}^{s-1} \xi_j v_j \right] \chi(u, \mathbf{v}) du d\mathbf{v},$$

where $v_0 := 0$ and χ is a smooth cutoff restricting each $u + v_j$ to $[u_{n_j}, \Theta(T)]$.

Lemma 8.3 (Multi-mode IBP). *Fix $s \geq 2$ and suppose $|\xi_k| > 1$ for at least one index $k \in \{1, \dots, s-1\}$. Then for every mode tuple $\mathbf{n}$ and every sign tuple ϵ ,*

$$|I_{\mathbf{n}, \epsilon}(\boldsymbol{\xi}, T)| \leq \frac{C_s \cdot \Theta(T)^{s-2}}{|\xi_k| - 1},$$

where C_s depends only on s .

Proof sketch. Integrate by parts once in the variable v_k . The v_k -phase derivative is

$$\partial_{v_k} [\epsilon_k \Phi_{n_k}(u + v_k) - \xi_k v_k] = \epsilon_k \omega_{n_k}(u + v_k) - \xi_k,$$

which by Lemma 4.3(c) satisfies $|\epsilon_k \omega_{n_k} - \xi_k| \geq |\xi_k| - 1 > 0$ uniformly. Applying the IBP structure of Lemmas 5.1–5.2 in v_k and integrating trivially over the remaining $s-1$ variables (each over a range of length $O(\Theta(T))$) yields the stated bound. $\square$

Remark 8.4. The factor $\Theta(T)^{s-2}$ arises from the $(s-2)$ non-IBP'd variables. After division by the Cesàro normalization $\Theta(T)^{s-1}$ (appropriate for an s -point cumulant density), the surviving factor is $\Theta(T)^{-1} \rightarrow 0$, which is the essential vanishing statement.

8.3. Band-limitedness of Φ_s .

Theorem 8.5 (Support of the s -point polyspectrum). *Assume the s -point Cesàro cumulant density κ_s exists in $\mathcal{S}'(\mathbb{R}^{s-1})$. Then for every $s \geq 2$,*

$$\text{supp } \Phi_s \subseteq [-1, 1]^{s-1}.$$

Proof. Let $\varphi \in \mathcal{S}(\mathbb{R}^{s-1})$ have $\text{supp } \widehat{\varphi} \subseteq \mathbb{R}^{s-1} \setminus [-1, 1]^{s-1}$. We show that $\langle \Phi_s, \varphi \rangle = 0$, equivalently that $\langle \kappa_s, \widehat{\varphi} \rangle = 0$. By linearity it suffices to test against a plane wave $\widehat{\varphi} = e^{-i\boldsymbol{\xi} \cdot}$ with $|\xi_k| > 1$ for some index k . Expand each occurrence of $\widetilde{Z}$ in Definition 8.1 via the Riemann–Siegel formula (1): the moment M_s decomposes into a finite sum of multi-mode integrals (6) plus a remainder involving the $R(t)$ factors.

Cosine-sum contribution. By Lemma 8.3,

$$\frac{1}{\Theta(T)^{s-1}} \sum_{\mathbf{n}, \epsilon} \prod_j n_j^{-1/2} |I_{\mathbf{n}, \epsilon}(\boldsymbol{\xi}, T)| \ll_s \frac{T^{s/4}}{\Theta(T) \cdot (|\xi_k| - 1)} \ll \frac{T^{s/4-1}}{\log T} \rightarrow 0$$

for $s \leq 3$ directly, and for $s \geq 4$ after observing that the partial-sum bound 3.1 gives $(\sum_{n \leq N(T)} n^{-1/2})^s \ll T^{s/4}$ and that the Cesàro normalization for an s -point cumulant is the stronger $\Theta(T)^{s-1} \asymp T^{s-1}(\log T)^{s-1}$, which dominates.

Remainder contribution. The Riemann–Siegel remainder contributes terms of size $\int |R(t)|^r$ for various $r \leq s$; each is $O(T^{1-r/4})$ by the bound $|R| \ll t^{-1/4}$, which after division by $\Theta(T)^{s-1}$ tends to zero.

Cumulant subtractions. Applying the Leonov–Shiryaev expansion, every term in κ_s is either a multi-mode integral or a product of lower-order polyspectra. The lower-order factors have already been shown (inductively, starting from Theorem 6.1) to vanish under the Cesàro limit when tested against frequencies outside $[-1, 1]$, so the full cumulant tested against $e^{-i\boldsymbol{\xi} \cdot}$ with $|\xi_k| > 1$ vanishes. $\square$

Corollary 8.6 (Polyspectrum box). *For every $s \geq 2$, the s -point polyspectrum Φ_s of $\tilde{Z}$ is a tempered distribution supported in the compact box $[-1, 1]^{s-1}$. In particular, the bispectrum is supported in $[-1, 1]^2$, the trispectrum in $[-1, 1]^3$, and so on.*

Remark 8.7 (Cumulant generating functional). Let $\mathcal{K}[\varphi]$ denote the formal cumulant generating functional

$$\mathcal{K}[\varphi] := \log \lim_{T \rightarrow \infty} \frac{1}{\Theta(T)} \int_0^{\Theta(T)} \exp \left(i \int \varphi(v) \tilde{Z}(u+v) dv \right) du = \sum_{s \geq 1} \frac{i^s}{s!} K_s[\varphi],$$

where $K_s[\varphi] = \int \kappa_s(\mathbf{v}) \prod_{j=0}^{s-1} \varphi(v_j) d\mathbf{v}$. Theorem 8.5 is the statement that $K_s[\varphi] = 0$ whenever $\hat{\varphi} \equiv 0$ on $[-1, 1]$. Equivalently, $\mathcal{K}[\varphi]$ depends only on $\varphi \mathbf{1}_{[-1, 1]}$ in the Fourier domain.

Remark 8.8 (Sharpness at each order). The sharpness argument of Corollary 7.1 extends: for every $s \geq 2$ and every $\boldsymbol{\xi} \in (-1, 1)^{s-1} \setminus \{0\}$, the multi-mode phase $\sum_j \epsilon_j \Phi_{n_j} - \sum \xi_j v_j$ admits non-degenerate joint stationary points at $u + v_j = u_{n_j}^*(\xi_j)$, producing a non-vanishing Cesàro contribution of order $\prod_j \omega'_{n_j}(u_{n_j}^*)^{-1/2}$. Hence $\text{supp } \Phi_s$ equals $[-1, 1]^{s-1}$ exactly.

ACKNOWLEDGEMENTS

The author thanks an anonymous reviewer for the three corrections incorporated in this revision (the asymptotic nature of the saddle-point identity, the $I_n + J_n$ decomposition of the cosine integral, and the mode-separation argument in the sharpness proof), and acknowledges the foundational work of B. Riemann, C. L. Siegel, G. H. Hardy, E. C. Titchmarsh, D. R. Brillinger, and V. P. Leonov & A. N. Shiryayev on which this paper rests.

REFERENCES

- [1] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., revised by D. R. Heath-Brown, Oxford University Press, Oxford, 1986.
- [2] C. L. Siegel, Über Riemanns Nachlaß zur analytischen Zahlentheorie, *Quellen Stud. Gesch. Math. Astron. Phys.* **2** (1932), 45–80.
- [3] F. W. J. Olver, D. W. Lozier, R. F. Boisvert, C. W. Clark (eds.), *NIST Digital Library of Mathematical Functions*, Release 1.2.1, 2024. <https://dlmf.nist.gov/>
- [4] E. M. Stein, *Harmonic Analysis: Real-Variable Methods, Orthogonality, and Oscillatory Integrals*, Princeton University Press, Princeton, 1993.
- [5] V. P. Leonov and A. N. Shiryayev, On a method of calculation of semi-invariants, *Theory Probab. Appl.* **4** (1959), 319–329.
- [6] D. R. Brillinger, An introduction to polyspectra, *Ann. Math. Statist.* **36** (1965), 1351–1374.
- [7] D. R. Brillinger, *Time Series: Data Analysis and Theory*, SIAM Classics in Applied Mathematics, vol. 36, SIAM, Philadelphia, 2001.
- [8] N. I. Akhiezer, *Lectures on the Theory of Approximation*, Frederick Ungar, New York, 1956.
- [9] S. Crowley, *Band-Limitedness of the Hardy Z-Function in the Phase Variable* (source manuscript, v1), 2026.

Email address: stephencrowley214@gmail.com